Networks Sub-module Assignment

Answers for Part 2

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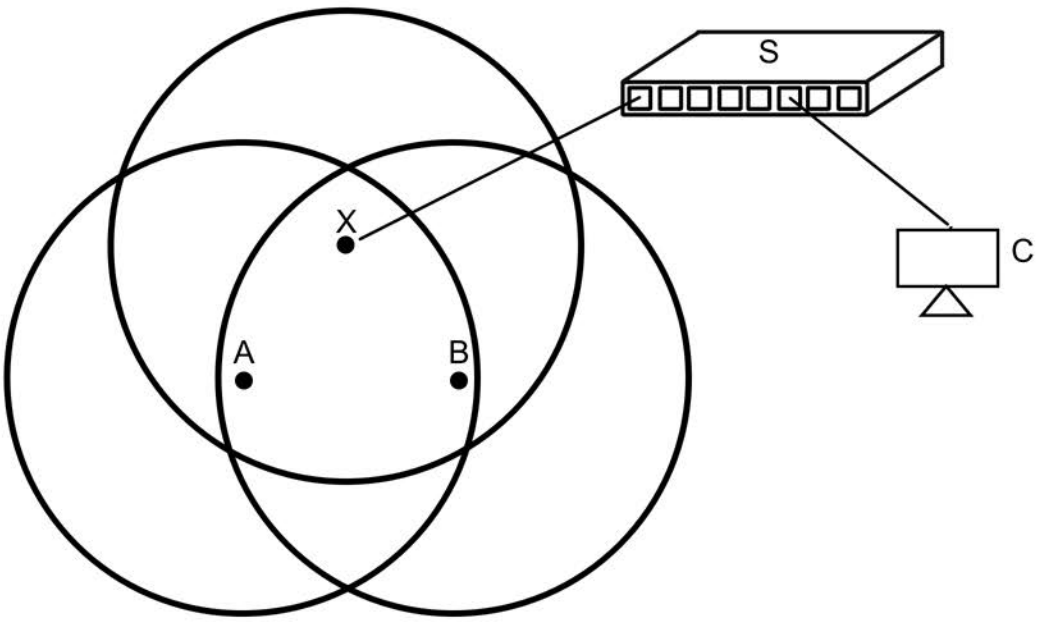
1. Sketch a topology to accurately reflect the connections of the network described above. Your topology should include all devices mentioned and their connections.



2. Which wireless user devices above can receive the frame sent by C? Why?

Both A and B can receive the frame sent by C. A starts sending its frame before C finishes sending, as the channel to X is idle, which makes the wireless channel busy. S will send C's frame to X at 50 μs where it will then repeatedly backoff and wait while A transmits to X. Once A finishes sending, B senses the channel as idle and sends its frame before X at 93 μs, X then gets its DIFS interrupted and continues to backoff until B finishes sending, allowing X to finally send C's frame. Once X is able to send, it'll be able to send to either A or B because both are under X's coverage and the channel is completely idle.

3. At what time does A start sending its frame (i.e., putting the frame on the transmission medium) to X? At what time does B start sending its frame to X? Explain.

A would start sending its frame to X at 25 µs, C is still sending its frame to S (finishing at 50 µs) and when A is ready to send at 20 µs, the wireless channel is idle for the 5 µs DIFS window meaning A is safe to start sending at 25 µs.

B would start sending its frame at 93 µs. Since A starts sending its frame before B is ready to send, this means the channel is busy and B has to back off repeatedly until A finishes. If A starts sending at 25 µs then A must finish sending at 85 µs. This means B must backoff 4 times, since B starts channel sensing at 40 µs this means B defers until 88 µs where it then channel senses for 5 more µs. Since the channel is idle, B starts sending at 93 µs.

4. Give the switching table of S at 60 µs. Explain.

MAC Address	Port
CC-CC-CC-CC-CC-CC	6

At time 60 µs, S only has the address of C, this is because C started and finished sending to S before 60 µs, this means S is able to add C to its switching table at 50 µs and then broadcast its frame across all ports. At 60 µs, this is the only frame S has received as A's frame is still in the process of sending and B is deferring. X also isn't added as S broadcasts C's frame through all its ports without logging the addresses.

5. If you connect a computer to port 2 of S, which frame(s) can you receive from all the above processes? Explain.

The new port 2 device would be able to receive C's frame. C's frame would be sent to S at 50 µs where it could then be broadcasted and sent to the new device. A and B's frames would have to be sent to S through X but since C's address is already on the switching table, the frame would be unicasted directly to C. If the destinations of each frame were changed to the new device then it would be able to receive all the frames as they would all be sent to S where they could then be forwarded to the new device.